

Project Statement: Blood Donation Matching System

Problem Statement

Blood donation coordination is still largely handled through scattered phone calls, WhatsApp forwards, and manual hospital registers. When a patient needs a transfusion urgently, there is no fast, reliable way to check which registered donors are both blood-type compatible and currently eligible to donate (donors must wait roughly 90 days between donations). This delay can be critical in emergencies, and manual tracking is prone to errors such as contacting an ineligible donor or losing track of who has already responded.

There is a need for a lightweight, dependable computational system that can validate blood-type compatibility instantly, track donor eligibility over time, and safely record and reverse donation transactions without data corruption or crashes.

Scope of the Project

The scope of this project is bounded to a lightweight, console-driven, in-memory matching application. The system manages donor records, blood requests, and donation transactions using Python dictionary indexing instead of persistent file systems or external database servers. The codebase covers donor registration, request intake, a compatibility-and-eligibility based matching engine, transaction-ID generation for completed donations, and reversible cancellation logic. It is tailored specifically for university grading and validation, not for deployment as a live medical system.

Target Users

- **Blood Bank Coordinators / Administrators:** Staff who register donors, log incoming blood requests from hospitals or individuals, and monitor the eligible donor pool by blood type.
- **Requesters (Patients / Hospitals):** Users who need to find compatible, eligible donors quickly and confirm a donation against a specific request.

High-Level Features

- **Donor Registration Module:** Captures donor identity, blood type, location, and last donation date, validating blood type against the eight standard types.

- **Compatibility-Based Matching Engine:** Cross-references a request's required blood type against the full donor-to-recipient compatibility table (e.g., O- as universal donor, AB+ as universal recipient) and filters by current eligibility.
- **Eligibility Tracking:** Computes donor eligibility using a rolling cooldown window (90 days) since the last recorded donation, rather than a simple boolean flag.
- **Transaction Reference Generator:** Produces a unique 8-character transaction ID for every completed donation match, linking a request to the donor who fulfilled it.
- **Reversible Cancellation Routine:** Safely reverses a completed match — reopening the request and restoring the donor's prior eligibility state — without leaving orphaned records.